

This could also be written in terms of the average transaction price P_{avg} as follows:

$$Expanded\ IS = \underbrace{S(P_0 - P_d)}_{Delay\ Related} + \underbrace{\left(\sum s_j\right)(P_{avg} - P_0)}_{Trading\ Related} + \underbrace{\left(S - \sum s_j\right)(P_n - P_0)}_{Opportunity\ Cost} + fees$$

This is the expanded implementation shortfall metric proposed by Wayne Wagner that makes a distinction between the investment and trading horizons. It was first identified in Wagner (1975) and later explained in Wagner (1991) and Wagner and Edwards (1993). The delay related component has also been referred to as the investment related cost. The delay cost component could be caused by the portfolio manager, buy-side trader, or broker-dealer. For example, see Almgren and Chriss (2000), Kissell and Glantz (2003), or Rakhlin and Sofianos (2006).

Example: A manager decides to purchase 5000 shares of a stock at \$10. By the time the order is finally released to the market the stock price has increased to \$10.25. If the manager is only able to execute 4000 shares at an average price of \$10.50 and the stock price at the end of trading is \$11.00 what is the expanded implementation shortfall cost by components? Assume total commission cost is \$80.

The calculation of the expanded implementation shortfall is:

$$\begin{aligned} Expanded\ IS = & \underbrace{5000 \cdot (\$10.25 - \$10.00)}_{Delay\ Related} + \underbrace{4000 \cdot (\$10.50 - \$10.25)}_{Trading\ Related} \\ & + \underbrace{1000 \cdot (\$11.00 - \$10.25)}_{Opportunity\ Cost} + \$80 = \$3080 \end{aligned}$$

The delay related component is: \$1250

The trading related component is: \$1000

The opportunity cost component is: \$750

Fixed fee amount is: \$80

Total expanded implementation shortfall = \$3080.

Notice that Wagner's expanded IS cost is the same value as Perold's IS. However, the opportunity cost in this example is \$750 compared to \$1000 previously. The reason for this difference is that the expanded IS measures opportunity cost from the time the order was released to the market as opposed to the time of the manager's decision. In actuality, the delay related